University of Science and Technology of Oran – Mohamed Boudiaf (USTOMB)
Faculty of Mathematics and Informatics
Computer Science Department

Master's Degree Courses

Artificial Intelligence and Applications

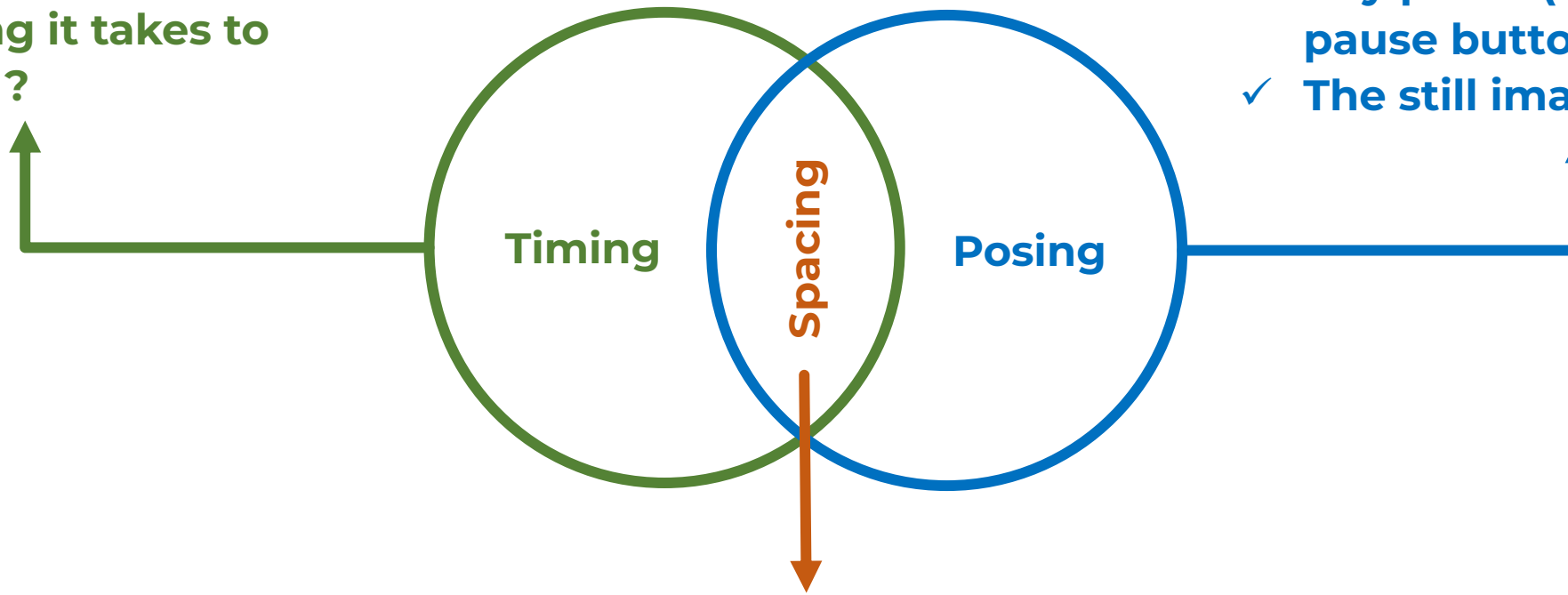
Animated Multimedia Applications Design

Animation technique principles

ebarki informatics

Building animations

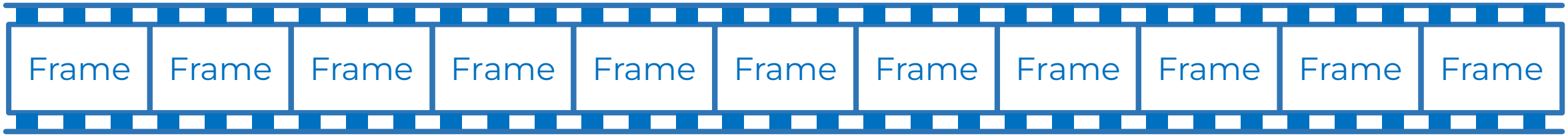
- ✓ How long it takes to happen ?
- ✓ Key poses (Pressing the pause button)
- ✓ The still image !



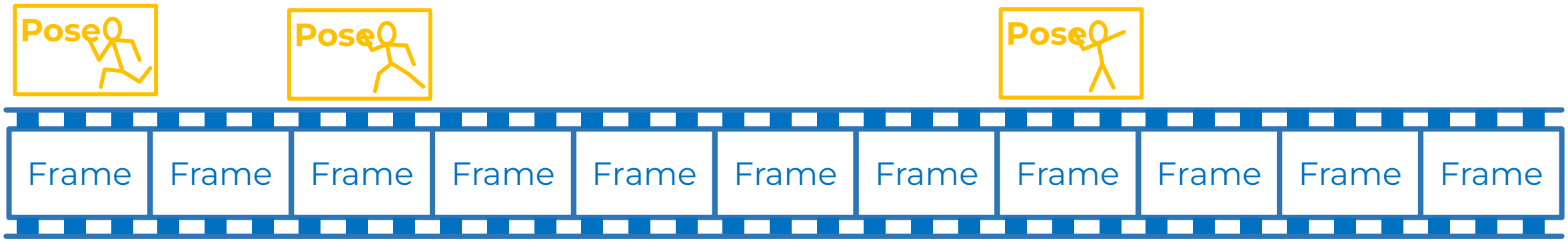
- ✓ Relationship between still images over time
- ✓ The space between the frames

Animation keys

- ✓ **Frames**
  - ❑ **24 or 30 frames per seconds**



- ✓ **Poses**
  - ❑ **Shots in Frame by Frame animations**
  - ❑ **Keyframes in automated generated animations**
  - ❑ **Keys, extremes, breakdowns, inbetweens**



Animation keys

- ✓ **Speed**
  - ❑ Seems fast when the poses are far apart
  - ❑ Seems slow when the poses are close together
- ✓ **Acceleration & Deceleration**
  - ❑ Speed up: Closer and closer together
  - ❑ Slow down: Further and further apart from each other
- ✓ **Squash & Stretch**
  - ❑ More realistic animations
  - ❑ Keep the object volume consistent
- ✓ **Anticipation**
  - ❑ Prepare the action
  - ❑ Highlight what will be animated
- ✓ **Overshoot**
  - ❑ Exaggerating or springiness
  - ❑ Follow through, overlapping, and drag actions
- ✓ **Continuity and curves**
  - ❑ Add flow and fluidity to the animation

Animation keys – Staging

We must guide the watcher towards the action



Animation keys – Staging

Includes Acting, Timing, View position, Setting



Animation keys – Staging

Introduce the actors one by one according to scenario order:

The viewer must not get confused about where to look !



Animation keys – Staging

Far away for actions, close up for facial expression



Animation keys – Staging

Main action in the middle or in the third of the screen



Animation keys – Staging

Main action clear an simple without being upstaged by other actions



Animation keys – Staging

Respect timing, one action at time !

No overlap ! Even put pauses !



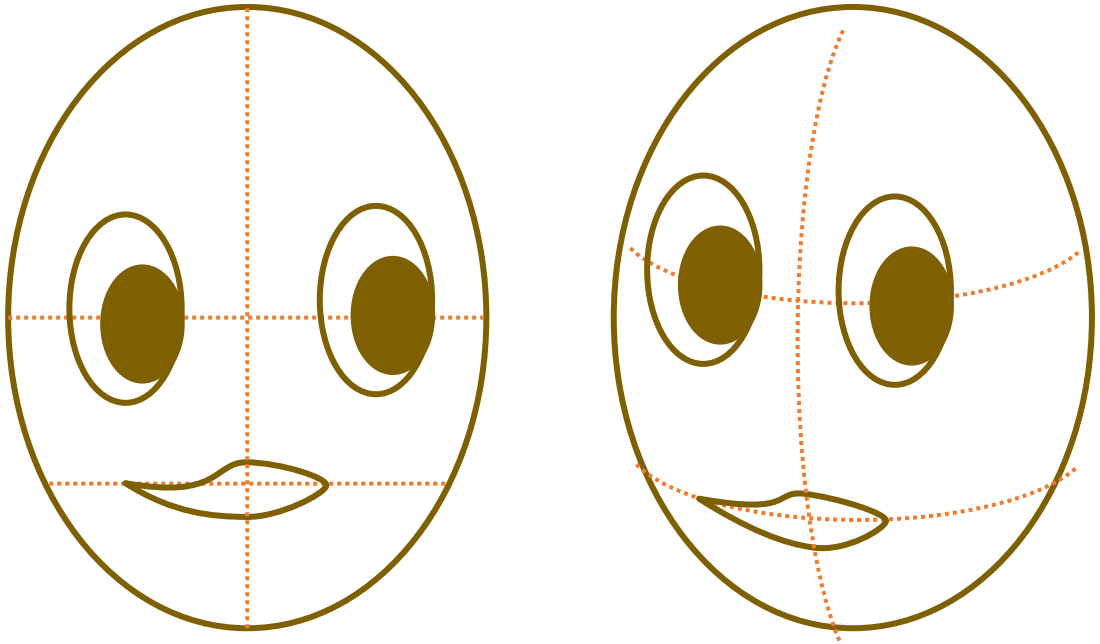
Animation keys – Staging

Keep displayed text on the

screen to be read three times !



Animation keys – Solid drawing



✓ Solid Drawing

- ❑ Give volume to 2D drawings

✓ Appeal (Using dynamic design)

- ❑ Variety of shapes,
- ❑ Proportions (Magnifying interesting parts),
- ❑ Keep simple (Symbolism for animating, details for illustrations)